

ThreatTrace AI: Intelligent Email Forensics & Threat Attribution Platform

Daily Engineering Work Log & Progress Report — Day 04

Sprint Phase: Sprint Phase 2: Core Forensic Engine Development | **Status:** Completed & Verified | **Schedule:** Day 04

1. Daily Sprint Overview and Context

On Day 4 of the development sprint, the engineering team executed primary development objectives for Cryptographic Authentication (SPF, DKIM, DMARC) & Domain Typo-Squatting Detection. Work focused on implementing production-ready algorithms, building asynchronous service endpoints, integrating network OSINT data feeds, and testing full-stack workflows across the extension, backend, and dashboard.

2. Daily Sprint Objectives & Scope

- 1) Implement and optimize core algorithms and services for Cryptographic Authentication (SPF, DKIM, DMARC) & Domain Typo-Squatting Detection.
- 2) Validate bidirectional JSON data contracts between backend FastAPI endpoints and React UI state stores.
- 3) Execute automated unit, regression, and performance test suites ensuring sub-200ms processing times.
- 4) Enforce strict data privacy and cryptographic chain-of-custody validation across all processed evidence.
- 5) Commit clean, verified, and documented source code artifacts into the main repository branch.

3. Subsystem Architecture & Domain Scope

- 1) **AI & Intent Classification Engine:** Heuristic keyword weighting, Bayesian text scoring, and brand impersonation detection.
- 2) **Email Header Forensics Core:** RFC 5322 MIME stream decoding, multi-hop Received latency math, and SPF/DKIM verification.
- 3) **IOC Extraction & Network OSINT:** Obfuscated URL unmasking, Punycode lookalike detection, and IP/ASN BGP enrichment.
- 4) **Threat Relationship Graph:** Bipartite graph data modeling, node serialization, and cross-incident campaign clustering.
- 5) **Risk Engine & Evidence Vault:** Composite Bayesian scoring, PostgreSQL/SQLite persistence, and Section 91 CrPC subpoenas.
- 6) **Forensic Cockpit & Extension UI:** React + Vite UI shell, Canvas graph physics, World Map, and Manifest V3 Gmail shield.

4. Individual Team Member Technical Contributions (Part I)

The following logs detail the specific engineering contributions, algorithmic implementations, and code deliverables produced by team members during Day 04:

1) Muniswami.Y — Lead AI & NLP Engineer

- a. **Domain Ownership:** Natural Language Processing & Threat Intent Analysis
- b. **Core Technical Work:** Executed specialized engineering tasks for Natural Language Processing & Threat Intent Analysis. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for Natural Language Processing & Threat Intent Analysis in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

2) Jeevan reddy.G — Backend & Email Forensics Lead

- a. **Domain Ownership:** MIME Parsing & Authentication Protocols
- b. **Core Technical Work:** Executed specialized engineering tasks for MIME Parsing & Authentication Protocols. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for MIME Parsing & Authentication Protocols in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

3) Lahari.K — Threat Intelligence & Network OSINT Lead

- a. **Domain Ownership:** IOC Extraction & Infrastructure Attribution
- b. **Core Technical Work:** Executed specialized engineering tasks for IOC Extraction & Infrastructure Attribution. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for IOC Extraction & Infrastructure Attribution in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

5. Individual Team Member Technical Contributions (Part II)

Detailed technical logs and code deliverables for systems architecture, graph intelligence, and frontend engineering:

4) Hemanth.P — Graph & Campaign Intelligence Lead

- a. **Domain Ownership:** Graph Data Modeling & Attack Clustering
- b. **Core Technical Work:** Executed specialized engineering tasks for Graph Data Modeling & Attack Clustering. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for Graph Data Modeling & Attack Clustering in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

5) Abbas.N — Systems Architect & Risk Engine Lead

- a. **Domain Ownership:** Risk Quantification & Backend Infrastructure
- b. **Core Technical Work:** Executed specialized engineering tasks for Risk Quantification & Backend Infrastructure. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for Risk Quantification & Backend Infrastructure in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

6) Moulaanbee.N — Frontend & Visualization Lead

- a. **Domain Ownership:** UI/UX Architecture & Browser Integration
- b. **Core Technical Work:** Executed specialized engineering tasks for UI/UX Architecture & Browser Integration. Implemented core logic, optimized algorithmic performance, and validated unit test assertions in alignment with Day 4 sprint milestones.
- c. **Key Tasks Executed:**
 - i. Developed core module logic and helper functions for UI/UX Architecture & Browser Integration in the active codebase.
 - i. Conducted code refactoring and memory profiling to ensure sub-200ms execution latency under load.
 - i. Authored automated unit tests with pytest covering edge cases, malformed payloads, and timeout handling.
 - i. Validated cross-module API contracts and committed clean, documented code artifacts to the repository.
- d. **Code Artifacts & Deliverables:** Module Source Code (`backend/app/services/`), Unit Test Suite, Technical Documentation.

6. System Data Flow and Module Integration

The system data pipeline operates across sequential stages to transform raw MIME email inputs into forensic intelligence:

- 1) **Client-Side Ingestion (Browser Extension):** Content scripts in `content.js` observe the Gmail DOM and extract raw RFC 822 MIME headers and text bodies upon user action.
- 2) **MIME & Received Header Traversal:** The FastAPI backend traverses the `Received` header chain in reverse order, filtering internal relays to identify the true public originating IP address.
- 3) **Cryptographic Authentication & DNS:** The asynchronous DNS engine queries SPF TXT records, DKIM public keys, and DMARC policies to verify cryptographic signature validity.
- 4) **IOC Extraction & Network Enrichment:** URLs are defanged and deobfuscated; originating IPs are enriched with GeoIP coordinates, ASN numbers, and threat intelligence feed lookups.
- 5) **Bayesian Risk Scoring & Graph Mapping:** The composite Bayesian risk formula calculates a 0-100 score while the Graph Engine connects entities into the Threat Relationship Graph.
- 6) **Real-Time UI Dispatch:** Enriched telemetry is broadcast to the React Forensic Cockpit and Cybercrime Portal queues for security analyst review.

7. Technical Challenges Faced and Engineering Solutions

1) Technical Challenge: Asynchronous Concurrency Contention

- a. **Problem Description:** Parallel network lookups during batch processing caused resource contention.
- b. **Root Cause Analysis:** Parallel network lookups during batch processing caused resource contention.
- c. **Engineering Solution Implemented:** Implemented asyncio semaphore pools with concurrency limits to ensure stable execution.

2) Technical Challenge: Malformed Input Stream Resilience

- a. **Problem Description:** Unusual or truncated email input strings caused decoding exceptions in edge cases.
- b. **Root Cause Analysis:** Unusual or truncated email input strings caused decoding exceptions in edge cases.
- c. **Engineering Solution Implemented:** Added defensive regex sanitizers and try-except recovery routines across all ingestion modules.

3) Technical Challenge: Component State Re-rendering Overhead

- a. **Problem Description:** Rapid telemetry updates triggered unnecessary UI redraws on the graph canvas.
- b. **Root Cause Analysis:** Rapid telemetry updates triggered unnecessary UI redraws on the graph canvas.
- c. **Engineering Solution Implemented:** Applied React.memo optimization and debounced state dispatches in frontend visualizer components.

8. Quality Assurance, Test Suite Metrics & Verification

Automated test suites were executed against Day 04 builds to verify RFC compliance and sub-200ms processing latencies:

- 1) **Test Case TC-04-01 (Module Unit Test):** Input: *Day 4 Core Algorithm Inputs* → Expected: *100% Function Assertion Pass* → **Status: PASS**
- 2) **Test Case TC-04-02 (API Endpoint Contract):** Input: *FastAPI JSON Controller* → Expected: *Clean HTTP 200 with Valid Schema* → **Status: PASS**
- 3) **Test Case TC-04-03 (Cryptographic Hash Audit):** Input: *SHA-256 Digest Verification* → Expected: *Exact Match Across Manifest Records* → **Status: PASS**
- 4) **Test Case TC-04-04 (Latency Benchmark):** Input: *50 Concurrent Synthetic Requests* → Expected: *Average Response Latency < 150ms* → **Status: PASS**
- 5) **Test Case TC-04-05 (Frontend Component Mount):** Input: *React 18 Dashboard Views* → Expected: *Rendered Cleanly with Zero Warnings* → **Status: PASS**

9. Security, Data Privacy and Chain-of-Custody Compliance

- 1) **Zero-Log Ephemeral Processing:** Raw email content is analyzed in volatile memory and is never permanently written to disk without explicit user reporting.
- 2) **Cryptographic Evidence Hashing:** All extracted email headers, body texts, and attachment metadata are hashed using canonical SHA-256 for court-admissible chain of custody.
- 3) **SSRF Protection:** Network URL deobfuscators strictly block loopback, private RFC 1918 subnets, and cloud metadata IP addresses (169.254.169.254).

10. Daily Milestone Verification & Next Day Schedule

- 1) **Day 04 Milestone Status:** 100% of scheduled tasks, API endpoints, and test suites completed and verified.
- 2) **Planned Objectives for Day 5:** Execution of subsequent sprint roadmap items, feature integration, and system hardening.
- 3) **Repository Synchronization:** All tested code committed and pushed to main Git repository.